AI-Powered Intelligent Talent Acquisition System

COMMON INTEGRATED SOLUTION REPORT

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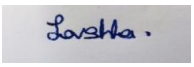
Sri Lanka

October 2026

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ABSTRACT

Recruitment still splits one hiring decision across a keyword screen, an unexplained match score, a subjective interview, and little evidence about whether a suitable candidate is likely to stay. This common report presents the AI-Powered Intelligent Talent Acquisition System (Group R26-IT-082), delivered as the Lanka Talent Insights workspace. The system is one containerised product with four analytical components that share candidate and vacancy records and return separate evidence. Resume intelligence extracts PDF, DOCX, and TXT curricula vitae and estimates skill proficiency with a multi-task DeBERTa-v3 model. Job candidate matching scores a vacancy against that profile with a local sequence classifier and can attach a grounded explanation. Interview evaluation transcribes a LiveKit session locally, reconstructs question–answer pairs, and scores technical answers with a calibrated DeBERTa short-answer model while reporting speech delivery separately. Hiring-risk support estimates attrition with a production CatBoost model and a Sri Lankan logistic regression line, and lets a recruiter test offer scenarios. Reported component results include skill-strength classification accuracy of 98.98 percent, CatBoost ROC-AUC of 0.807, Sri Lankan hold-out ROC-AUC of 0.903, and public short-answer quadratic weighted kappa of 0.758. Interview-domain agreement is preliminary, and labelled ranking metrics for matching are still outstanding. The platform supports the recruiter. It does not hire or reject on its own.

Keywords: talent acquisition, explainable artificial intelligence, resume proficiency, interview scoring, attrition risk

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial intelligence
API	Application programming interface
ATS	Applicant tracking system
MAE	Mean absolute error
MRR	Mean reciprocal rank
NDCG	Normalised discounted cumulative gain
NLI	Natural language inference
NLP	Natural language processing
QWK	Quadratic weighted kappa
ROC-AUC	Area under the receiver operating characteristic curve
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
STT	Speech to text
XAI	Explainable artificial intelligence

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Hiring asks four different questions: whether a curriculum vitae contains credible skill evidence, whether that evidence fits a vacancy, whether the candidate can explain the work, and whether the person is likely to remain after an offer. Applicant tracking systems accelerated document storage and keyword filters. They did less for proficiency, explanation, interview consistency, and pre-offer retention risk [14], [15]. A fluent file can hide thin project evidence. A similar paragraph can still contradict a requirement. A technically correct spoken answer can be marked down for hesitation. A strong match can still be a poor retention bet. These are different errors. One automated hire-or-reject label would hide which error occurred. Group R26-IT-082, the AI-Powered Intelligent Talent Acquisition System, treats them as four linked subproblems inside one product. The implemented product is the Lanka Talent Insights workspace. Candidates keep a profile, upload a curriculum vitae, apply to open vacancies, and join a scheduled interview. Recruiters publish vacancies, read a versioned match score, open a skill analysis, run a LiveKit interview, review a question-level assessment, and request an attrition estimate with a What-If offer scenario. A React and Vite client talks to an Express gateway. MongoDB stores profiles, vacancies, applications, and interviews in the database lanka_talent. Sri Lankan survey evidence is used on the retention side. Public and adapted benchmarks are used where local outcome labels do not yet exist. Full experimental detail stays in the four individual theses.

1.2 Stakeholders and user needs

Primary users are recruiters who must shortlist with limited time, and candidates who must see how their documents and interviews are read. The organisation carries the cost of a poor hire or an early exit. The four theses agree on one user need: each automated output should be inspectable, and the offer should remain a human decision [14]. Recruiters need a skill profile, an explained match, a question-level interview record, and a retention view that can be recomputed when pay, work model, scope, or manager conditions change. Candidates need one account for the 1 profile, the

application, and the interview. A formal multi-recruiter acceptance study of the whole workspace is still open. The interview pilot of ten items checks scoring validity. It is not a usability study of the dossier.

1.3 Alignment with the Sustainable Development Goals

The project aligns with three United Nations Sustainable Development Goals [11]. Goal 8, decent work, is relevant because a clearer screen can surface candidates whose skills are worded differently from the vacancy. Goal 9, industry and innovation, is relevant because the contribution is a working system that joins language models, local speech transcription, and tabular risk models. Goal 10, reduced inequalities, is relevant because hiring models can reproduce bias in curricula vitae, job text, and past attrition labels [14]. The design response is to keep four scores separate, to show evidence, and to leave the decision with the recruiter. These are plausible contributions. The project has not measured employment, discrimination, or retention outcomes in a live organisation. Section 7.5 states the conditions for any stronger claim.

1.4 Purpose and scope of the common report

This report describes the integrated problem, the four-component solution, the way the components share data, the implementation of that sharing in the repository, and the group-level evidence available in October 2026. It follows the common-report structure required for the module: context, problem and subproblems, solution structure, integration, implementation, evaluation, commercialisation, and conclusion. Detailed model recipes, hyperparameter tables, and dataset construction notes remain in the individual theses of Jayantha C. H. (IT22226914), Deshan M. A. K. (IT22195616), Tennakoon T. M. S. S. (IT22197832), and Fonseka W. G. K. (IT22248794). The report is written against the implemented source, not only against the intended design. Where an individual thesis marks a number as illustrative, this report does not repeat that number as a measured result. Where a Topic Assessment Form item was not built, this report names it as future work. Certification authenticity checking 2 is in that category. It was named beside interview evaluation and was not part of the frozen scoring model.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

The common problem is that technical recruitment decisions are made from four kinds of evidence that current tools either omit or collapse. Recruiters need a structured reading of the curriculum vitae, a transparent comparison with a vacancy, an auditable reading of the spoken interview, and an estimate of early leaving risk that can be inspected before an offer. They need these readings in one workflow, with enough information to disagree with any one of them. The problem is the missing joint decision-support record. It is not the absence of a particular model family. The problem is socio-technical as well as algorithmic. A score that cannot be contested is a poor fit for a decision that affects employment [14]. A system that mixes speaking style into a technical mark, or that hides a failed model call behind another component's number, would make the record less trustworthy than a manual file. The solution boundary follows from that: the platform recommends and explains, and the recruiter decides.

2.2 Root causes and overall research gap

Four causes recur across the individual literature reviews and are visible in the product gap the group set out to close. First, applicant tracking practice still leans on keyword presence. Contextual encoders such as BERT and DeBERTa show that wording and token role both matter, which is why a skill alias or a paraphrased answer can be missed by a term list [1], [2], [3]. Second, similarity is not the same as agreement. Sentence embeddings are a useful baseline for ranking text, and they can still assign a high score to a fluent contradiction [4]. Third, interview products often store video or a transcript and then return either no score or a score without an account of coverage and contradiction [5]. Fourth, attrition models are usually built on current employees and return a passive probability. They rarely sit beside a live application, and they rarely let a recruiter test whether a change in pay or work model moves the estimate [6], [7]. 4 Job–person matching itself is an established recommendation problem [9]. What is less common in one operational system is the joint presence of proficiency estimation, an explained match, a reference-conditioned interview score, and a pre offer retention scenario, with each output stored separately. Commercial video interview and coding

platforms cover scheduling or task scoring. They do not, in the form reviewed by the group, publish this four-part local pipeline. The overarching gap is therefore an integrated, inspectable hiring record for technical roles, with Sri Lankan survey evidence used on the retention side where a local sample exists, and with public or adapted benchmarks used where local outcome labels do not yet exist. The novelty claim of the group is the integration and the decision boundary, together with component methods that are specific enough to evaluate. Transformers, gradient boosting, and speech recognition are prior work. The project's claim is that these methods can be bound to one candidate and one vacancy without collapsing their outputs, and that failure of any one model remains visible.

2.3 Subproblem and component mapping

The common problem separates into four subproblems. Each is necessary. A match score computed on an unparsed document inherits layout noise. An interview score computed without a vacancy still needs the application context in which the recruiter will read it. An attrition estimate that ignores the match and the interview omits signals the recruiter already has. None of the four, alone, answers whether the person can do the work, can explain it, and is likely to stay. Table 1 maps the subproblems to owners and to the services in the repository.

Table 1: Subproblem and component mapping

Subproblem	Owner	Repository services
Heterogeneous CVs hide proficiency	Jayantha C. H. IT22226914	cv-extraction-service, cv-profile-analysis-service, resume-strength-model-service
Match scores lack an explanation	Deshan M. A. K. IT2219516	cv-matching-service, shared explanation service
Spoken interviews are scored inconsistently	Tennakoon T. M. S. S. IT22197832	speech-to-text-service, interview-analysis-service, interview-answer-model-service, LiveKit via job-service
Suitability is not retention risk	Fonseka W. G. K. IT22248794	attrition-service, attrition-model-service, early-attrition-model-service

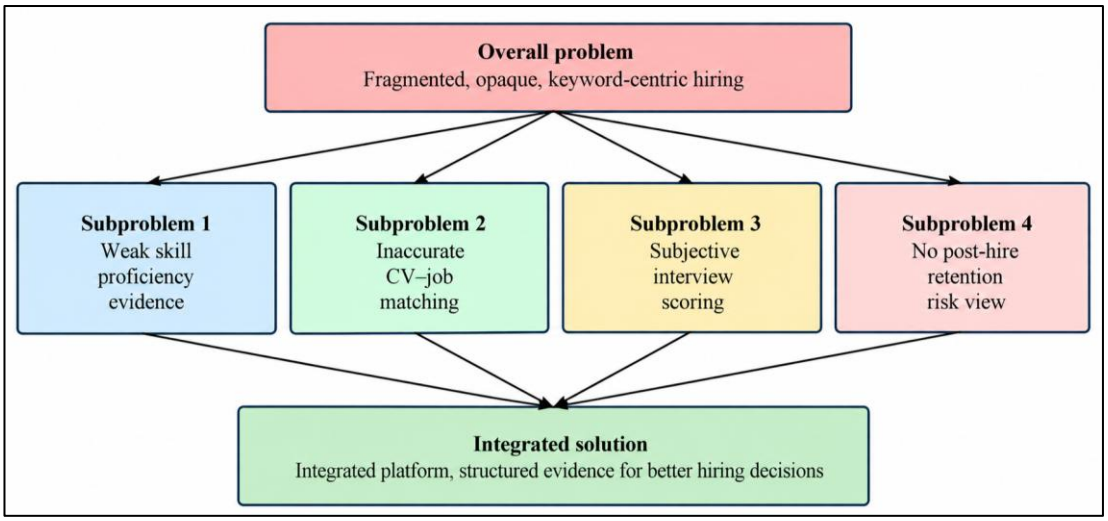


Figure 1: Four interdependent subproblems of the hiring gap

2.4 Overall aim and group objectives

The group aim is to design, implement, and evaluate one human-in-the-loop talent acquisition system that presents skill evidence, vacancy match evidence, interview evidence, and pre-offer attrition evidence on the same candidate and vacancy, without issuing an automatic hiring decision. Table 2 states the objectives in a form that Section 6 can mark as achieved, partly achieved, or not achieved.

Table 2:Group objectives and the evidence used to judge them

ID	Objective	Evidence
GO1	Share one candidate, vacancy, and application record across the four analyses.	Gateway, candidate service, and job service; application payload includes match score and optional skill analysis.
GO2	Return four distinct evidence types and keep them in separate fields.	Match, analysis, and attrition responses are different contracts. No service writes a hire or reject label.
GO3	Make model failure visible instead of substituting another component's score.	Interview analysis records an unavailable scorer. Attrition returns model warnings and an agreement flag. Matching returns an error if the local model is not loaded.

GO4	Report component quality on the datasets each thesis defines, without mixing those datasets.	Held-out resume, attrition, and short-answer metrics, with the interview pilot and the matching ranking study reported at their true status.
GO5	Provide one recruiter path from application to interview to risk review.	Implemented pages for jobs, applications, skill analysis, interview scheduling, LiveKit room, analysis, and attrition prediction.

2.5 Scope assumptions and constraints

The complete solution covers English-dominant technical curricula vitae in PDF, DOCX, and TXT with selectable text, vacancy records created in the workspace, live or recorded interview audio transcribed locally, and attrition estimates from candidate, job, and simulation fields. Image-only scans that need optical character recognition are outside the current extraction service. The interview scorer is specified for English software-engineering answers. Facial emotion, personality, and honesty are not inferred. Certificate authenticity checking is named in the topic scope of the interview component and is not an implemented function. Assumptions are that users authenticate before uploading personal documents, that a recruiter will read low-confidence or needs-review items, and that model outputs are decision support. Constraints are local model files rather than a runtime download from a public model hub for the matching, resume-strength, interview-scoring, and speech services; a Gemini call only on the explanation path; and a single MongoDB 6 deployment for the current configuration. Prospective validation against later employment outcomes is required before any claim that the attrition score predicts real exits.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

Lanka Talent Insights is one system. The candidate-facing side collects identity, a curriculum vitae, and applications. The recruiter-facing side publishes vacancies and reads a growing application dossier. The dossier can contain a parsed skill profile, a match probability and classification, an interview transcript and question-level assessment, and an attrition result with explanations. The value of the complete solution is that these artefacts refer to the same person and the same role, and that a weakness in one artefact does not silently overwrite another. The central outcome for a recruiter is a shortlist that can be defended. For a given application the recruiter can see why a skill was graded, why the match was marked suitable or in need of review, how each interview answer compared with a reference, and which conditions move the attrition estimate. The central outcome for a candidate is a single account in which the application and the interview appointment live together. Neither outcome is an automated employment decision.

3.2 High level architecture

Figure 2 shows the deployment boundaries taken from docker-compose.yml and from the gateway route table. The browser talks only to the gateway on port 8080. The gateway checks a JSON Web Token, enforces administrator routes, and forwards JSON or multipart bodies to the owning service. Platform services cover authentication, candidate profiles, and jobs. Analytical services are grouped by component. A shared explanation service on port 4012 calls Gemini when an explanation is requested. Model weights for matching, resume strength, interview scoring, natural language inference, and speech are loaded from local directories.

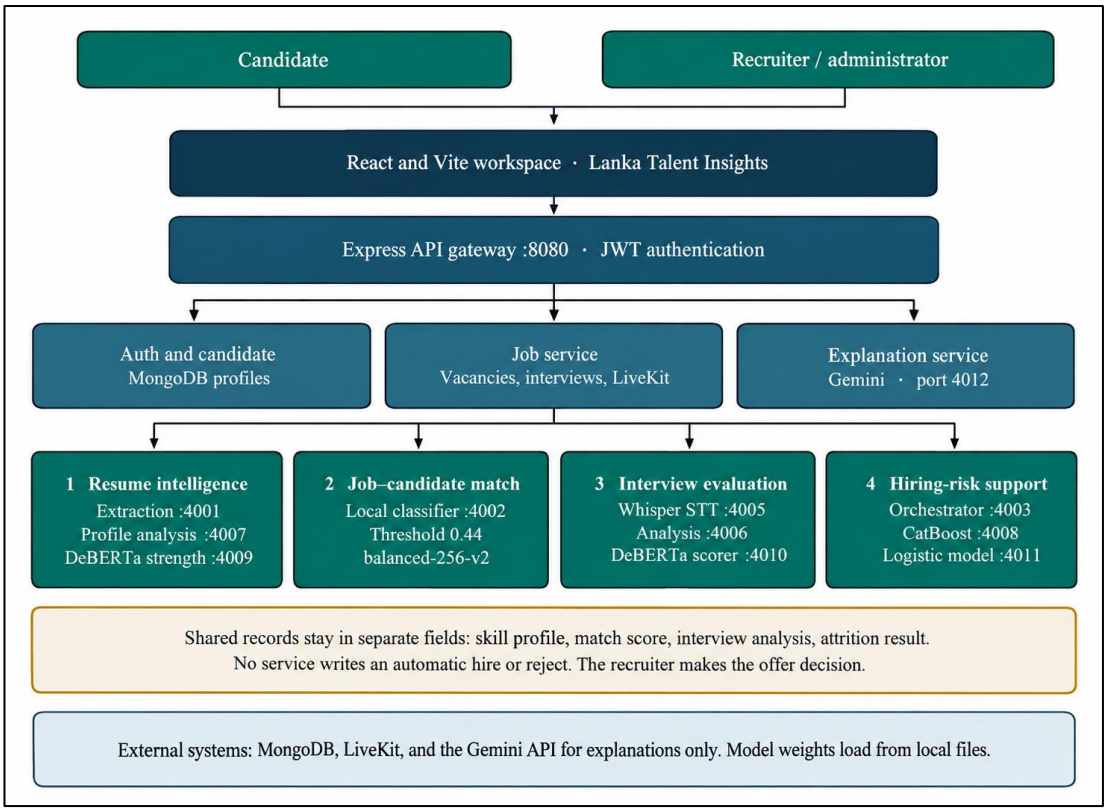


Figure 2:High-level architecture of the integrated talent acquisition system

The architecture is polyglot and containerised. Node.js and Express implement the gateway, authentication, candidate, job, extraction, matching, attrition orchestration, and profile-analysis services. Python services host CatBoost, the early-attrition logistic model, the resume-strength DeBERTa model, and the interview-answer DeBERTa model. MongoDB is the system of record. LiveKit provides the interview room tokens issued by the job service. This split keeps a heavy model process from blocking document storage, and it lets one model be restarted without taking down vacancies or accounts.

3.3 Component 1 summary

Jayantha C. H. (IT22226914) owns resume intelligence. The gateway accepts PDF, DOCX, and TXT. Extraction returns text. Profile analysis parses sections, detects skills against an alias-aware catalogue, and calls a fine-tuned multi-task DeBERTa v3 model. The model predicts project strength, skill-evidence strength, and experience–project alignment, each as a 0–100 score and as Beginner, Intermediate, or Advanced.

Gemini then grounds those scores in curriculum vitae evidence. The 9 profile is stored for the candidate and can be attached to an application. If the model service is down, deterministic parsing remains and a neural score is not invented. Held-out metrics are reported in Section 6.3.

3.4 Component 2 summary

Deshan M. A. K. (IT22195616) owns job–candidate matching. Input is a job object and a candidate object. The matching service builds a job text and a candidate text, compacts each side, and scores the pair with a local sequence-classification model loaded through a transformers runtime that does not fetch remote weights. The model identifier in the deployment configuration is cv-match-browser-local. Token length is capped at 256. The deployed decision threshold is approximately 0.440. A probability at or above the threshold is labelled Suitable Match; otherwise the label is Needs Review. The response also carries a percentage, the model identifier, and the input version balanced-256-v2. Explanation is a second call. It requires the raw curriculum vitae text, the candidate, the job, and the match result, and it is produced by the shared Gemini explanation service. The gateway scores a candidate when an application is submitted and can backfill older applications whose stored score uses an earlier input version. The individual matching thesis defines a ranking evaluation using Precision@K, recall at K, mean reciprocal rank, and normalised discounted cumulative gain, and it compares a proposed matcher with keyword and TF–IDF baselines. The thesis text states that the numerical tables prepared for that comparison are illustrative and require a labelled experiment before they are reported as findings. This common report therefore describes the implemented scorer and the evaluation design, and it does not adopt those illustrative ranking figures as group results.

3.5 Component 3 summary

Tennakoon T. M. S. S. (IT22197832) owns interview evaluation. The job service issues a LiveKit token for a scheduled interview. Audio is transcribed by a local Whisper-family model and is not sent to an external speech API for scoring [5]. The analysis service pairs interviewer questions with the following candidate turns. Unanswered questions are stored as unanswered and receive no artificial mark.

Answered questions are scored by the frozen model Final_ASAG_Interview_Scorer_V5, a multi-task DeBERTa-v3 short-answer grader. It fuses the encoder state with eight lexical coverage features and three frozen natural-language-inference probabilities, then applies saved calibration to a 0–100 score and a class of Wrong, Poor, Average, Good, or Excellent [2], [3], [16]. A parallel inference view reports coverage and contradiction. Fillers, latency, duration, and word count are shown separately and do not change the technical score. The expert rubric follows Bloom's revised taxonomy [10]. Public-set and pilot numbers are in Section 6.3. Certificate authenticity checking is future work.

3.6 Component 4 summary

Fonseka W. G. K. (IT22248794) owns hiring-risk decision support. The orchestrator accepts a candidate and a simulation, calls two model services, and records whether they agree. The study benchmarked CatBoost against logistic regression, XGBoost, LightGBM, and random forests under class imbalance [7], [8]. The deployed classifier is CatBoost, attrition-risk-catboost-v7, at threshold 0.33, and it can return TreeSHAP contributions [6]. A logistic service, early-attrition-logistic-v1, scores eleven early-risk features covering pay, job switching, match, similarity, interview, notice, location and work model, training, mentorship, career development, and certification. If both model URLs are unset, a weighted rule is used and labelled rule based. What-If edits travel in the simulation and do not overwrite the stored candidate. The adapted 1,470-row line and the 835-response Sri Lankan line answer different questions. Their metrics are in Section 6.3. A prospective employer study is still required.

3.7 End to end user and data flow

Figure 3 shows the path a successful application follows. The candidate uploads a curriculum vitae. Extraction and profile analysis produce a skill profile on the candidate record. When the candidate applies, the gateway loads the vacancy, calls the matching service, optionally attaches the stored skill analysis, and posts the application to the job service with the match score. An administrator can open the application, request a fuller curriculum vitae analysis, schedule an interview, and later request interview analysis from the transcript. Attrition prediction is a further 11 call that can

include the match and interview context and a simulation of offer conditions. The recruiter then reads the dossier. Nothing in this path writes an employed or rejected state by itself. Status changes on the application are explicit administrator actions.

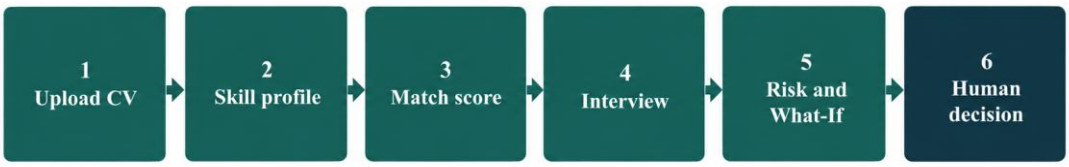


Figure 3: End-to-end path from curriculum vitae upload to a human hiring decision

Data identity is shared through user identifiers, job identifiers, and application identifiers. The match score stored on the application includes probability, percentage, threshold, classification, model identifier, and input version. Interview analysis is stored by interview identifier in the analysis database. Attrition is computed on demand from the candidate snapshot and the simulation, so a What-If run can be compared with the baseline without destroying it. This is the cohesive operation the four components were built to support.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

Integration is interface-first and runs through one gateway, which is the only browser-facing process. Each analytical owner exposes a small HTTP contract. Platform records live in the candidate and job services. Analysis services may use their own database, and they key results by the same user, job, or interview identifiers. Parsing and matching sit on the application path and must fail clearly. Interview analysis runs after a transcript exists and is limited to administrators. Attrition can run on a candidate snapshot and a simulation before or after the interview. Risk is managed by timeouts, by storing the model and input version beside a match score, and by refusing to present a fallback narrative as the frozen model score.

4.2 Interfaces data contracts and shared models

Table 3 lists the contracts that bind the components. Appendix A extends the list to the gateway routes a reviewer can trace in proxyRoutes.js. Validation is strict on the analytical calls: matching requires job and candidate objects and sufficient candidate evidence; explanation requires raw text; attrition requires a candidate object and, at the gateway, a simulation object; interview analysis is an administrator action on an existing interview.

Table 3: Principal interfaces and data contracts

Contract	Request	Response used by the dossier
POST /api/cv/extract	PDF, DOCX, or TXT file	Extracted text for parsing
POST /api/cv/analyze	Candidate object, optional raw text and job	Skill profile and proficiency explanation
POST /api/match/score	Job and candidate objects	Probability, percentage, threshold, Suitable Match or Needs Review, model, input version
POST /api/match/explain	Raw text, candidate, job, match result	Grounded match explanation
POST /api/interviews/:id/analysis	Optional reference answers	Question-level scores, ratings, NLI evidence, speech indicators, model status

POST /api/attrition/predict	Candidate and simulation	Primary risk, both model outputs, agreement, warnings, explanation

Shared identifiers are the authenticated user subject, the job identifier, the application identifier, and the interview identifier. The match input version balanced-256-v2 is part of the contract so that a later model change does not leave silent stale scores. 13 The gateway backfills an application when the stored version differs and both the candidate and the job are still available. There is no shared neural model across components. DeBERTa weights for resume strength and for interview scoring are different artefacts. CatBoost and the logistic attrition model are different artefacts. Explanations share a Gemini client, which is a language layer over evidence already computed, not a fifth scorer.

4.3 Shared nonfunctional requirements

Table 4 states the requirements that apply to the whole product. They are implemented to different depths. Authentication, role checks, upload limits, local inference, and separation of evidence are present in the repository. Formal load tests, an accessibility audit, and a completed privacy-impact assessment are not yet reported as group results and are listed as release conditions in Chapter 7.

Table 4:Shared non-functional requirements

Quality	Requirement for the integrated system	Current provision
Security	Authenticated access; administrator actions separated from candidate actions.	JWT on the gateway; requireRole(admin) on vacancies, interviews, and analysis.
Privacy	CV text, audio, and scores are personal data. Audio for scoring stays local.	Local Whisper path; Gemini used for explanations. PDPA duties apply before a live pilot [12].
Reliability	A failed model call must be visible.	503 or recorded unavailable status; attrition modelWarnings; rule-based method labelled when used.

Integrity	Scores from different models must not overwrite each other.	Separate fields and response objects, as in Table 4.1.
Maintainability	Owners can change a model without rewriting the workspace.	One service per model family; input version stored on the match.
Usability	A recruiter can read evidence without opening four tools.	Single workspace pages for analysis, matching, interviews, and risk.
Fairness	Delivery style and demographics are not silent technical penalties.	Interview speech metrics are separate. The scorer does not receive age or gender. Attrition remains advisory [14].

4.4 Integrated workflow and exception handling

In normal operation the application is created only after a match score returns. Skill analysis is attached when the candidate already has one. Interview analysis runs on an ordered transcript. Attrition prediction runs the CatBoost service and the early attrition service together and reports modelAgreement as agreement, disagreement, or single-model. A Gemini explanation is added when that call succeeds. The important failure paths are part of the design. An unsupported file is rejected at the gateway with HTTP 415. A match call without a job title or candidate evidence is 14 rejected with HTTP 400. If the local matching model is not loaded, the service responds that the model is unavailable rather than returning a keyword score under the model's name. If the interview scorer cannot be called, the analysis record stores that failure and does not present a fallback analysis as the frozen DeBERTa score. If both attrition model services fail, the orchestrator returns HTTP 503. If only one succeeds, the response is marked single-model and the error string is kept in modelWarnings. Profile-photo enrichment is allowed to fail open so that a candidate service fault does not hide the application list. These behaviours are what make the four components safe to operate together. Table 5 summarises the split between the normal dossier and the recorded failures.

Table 5:Normal operation and the main failure paths

Path	What the recruiter should see
Normal application	Stored match score with model name and input version, plus skill analysis when it exists.
Matching model down	Error. No invented Suitable Match label.
Interview model down	Analysis saved with model status unavailable or partial. Fallback text is labelled as fallback.
One attrition model down	The surviving model, plus a warning and a single-model agreement flag.
Both attrition models down	Service unavailable. The rule-based score is used only when model URLs are not configured, and it is labelled rule-based.
Explanation service down	The numeric or class result remains. The missing explanation is not filled with an unsupported reason.

4.5 Team contribution and dependency matrix

Table 6 records the agreed split. Shared work is the gateway, authentication, the job and candidate records, container composition, and the explanation client. Each member's research design and model evaluation belong to that member's thesis. Dependencies run forward along the hiring path: matching and attrition read candidate and job fields that resume parsing helps to structure; interview analysis reads the interview created from an application; attrition can read match and interview signals as features, and those signals remain independently visible.

Table 6:Team contribution and dependency matrix

Member	Research ownership	Depends on	Provides to the others
Jayantha C. H.	Parsing, skill catalogue, multi-task proficiency model, CV explanations	Auth and candidate profile	Structured skills and evidence for matching, interview context, and review screens
Deshan M. A. K.	Pairwise match model, threshold, explanation request, score versioning	Job and candidate objects	Match score stored on every new application

Tennakoon T. M. S. S.	Transcription, question pairing, DeBERTa grader, NLI evidence, speech panel, expert pilot	Scheduled interview and transcript	Question-level assessment the recruiter can read beside the match
Fonseka W. G. K.	CatBoost and logistic lines, threshold, TreeSHAP, What-If, dual-model agreement	Candidate snapshot; optional match and interview features	Retention risk and scenario comparison before an offer

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

Major choices follow from the shape of the problem. A single-page workspace was required so that a recruiter would not leave the dossier to open a notebook. React and Vite provide that workspace, and LiveKit's browser client joins the interview room. An API gateway was required so that upload rules and roles would not be reimplemented in every service. MongoDB fits profile and vacancy documents whose fields grow as new analyses are attached. Python was used where the research models already lived: CatBoost, scikit-learn logistic regression, and PyTorch DeBERTa checkpoints. Node was used for I/O-bound orchestration. Local model files were chosen so that a demo or a viva does not depend on downloading weights at startup, and so that interview audio is not required for a cloud speech API. Gemini is confined to explanation text because the scores themselves must remain reproducible from local weights. Table 7 lists the stack.

Table 7:Common technology stack

Layer	Choice	Reason in this system
Client	React, Vite, React Router, LiveKit client	One workspace for profile, jobs, analysis, and the interview room
Edge	Express gateway, JWT, Multer	Auth, role checks, PDF/DOCX/TXT limit, audio limit
Records	MongoDB database lanka_talent	Profiles, jobs, applications, interviews, transcripts
Resume	Node extraction and analysis; Python DeBERTa service	Separate parsing from the strength model
Matching	Node service, local transformers classifier	Pairwise score with a stored input version
Interview	LiveKit, faster-whisper, Python DeBERTa and NLI	Local transcript and a frozen grader
Attrition	Node orchestrator, CatBoost, logistic regression	Two models, SHAP values, labelled fallback
Explanation	Node Gemini client	Natural-language reasons over computed evidence

Delivery	Docker Compose	One command brings up the integrated path

5.2 Development and integration process

Ownership follows Table 1 Members integrated through the gateway rather than through a shared scoring library, so one member's metric is not written into another member's response. The match input version shows the pattern: old scores stay identifiable, and the gateway can recompute them. Health routes report whether a model loaded. Each thesis separates modelling objectives from system objectives. A strong public-set score is still the wrong thing to show a recruiter if a failed call is hidden, which is why the implementation carries status fields and method labels.

5.3 Deployment and operational configuration

Docker Compose builds the frontend, the gateway, and the service set. The gateway receives service URLs for extraction (4001), matching (4002), attrition orchestration (4003), jobs (4004), speech (4005), interview analysis (4006), profile analysis (4007), the CatBoost service (4008), resume strength (4009), the interview scorer (4010), early attrition (4011), and explanations (4012). Authentication and the candidate service use 3001 and 3002. Secrets and the MongoDB URI are environment variables. MODEL_THRESHOLD The CatBoost 0.33 container and is started model with file attrition_risk_catboost_v7_optuna.joblib. The matching container is started with MATCHING_THRESHOLD 0.4399277865886688 and MATCHING_MODEL_ID cv-match-browser-local. The interview scorer is started from a local V5 directory and a local natural-language-inference directory. The speech container defaults to a small Whisper checkpoint on CPU with int8 compute, which is an operational choice for a student deployment. A larger checkpoint can be selected with WHISPER_MODEL_SIZE. That default should be stated in any demonstration, because transcription errors flow into the interview score and have not yet been measured as a word-error study. Backup of MongoDB is the operator's responsibility in the current compose file; a production deployment would add scheduled dumps and a restore drill before candidate data were accepted from outside the project team.

5.4 Evidence of team execution

Execution evidence is the repository: four analytical paths, a gateway that calls them in hiring order, candidate and administrator pages, model files named in the Compose configuration, and the four theses. Table 6 is the agreed ownership record. Activity logs are omitted, as the common-report template requests. The integrated path is implemented. Component studies are uneven: resume proficiency, 18 attrition, and public short-answer grading have held-out numbers; interview transfer is a pilot; matching rank metrics are not yet confirmed on labelled pairs. Chapter 6 uses that status directly.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

Group evaluation asks whether the four components can be used as one dossier, whether their failures stay visible, and whether each component's own study supports its role. It does not average a DeBERTa accuracy with a CatBoost ROC AUC. Those numbers answer different questions on different data, and mixing them would overstate the system. Table 8 sets the criteria before the results.

Table 8:Integrated evaluation questions and success criteria

Question	Success criterion	What would count as a miss
Can one application carry skill and match evidence?	Apply flow stores a versioned match score and can attach analysis.	Match is computed in a notebook that the application cannot read.
Can interview and attrition be opened on that application?	Administrator can analyse a transcript and request a prediction.	Either result can only be produced outside the workspace.
Are the four results distinct?	Separate contracts and no automatic hire label.	One blended score is the only stored outcome.
Are component claims traceable?	Each metric is tied to the dataset in the owning thesis.	A public-set kappa is described as a Sri Lankan hiring outcome.
Is failure explicit?	Down models produce errors, warnings, or an unavailable status.	A keyword or rule score is labelled with the neural model name.

6.2 Integration and end to end testing

Integration testing is the set of gateway behaviours a reviewer can execute. File type and size are checked before extraction. Job and candidate bodies are checked before matching. An application post loads the vacancy, scores the pair, and writes matchScore. Administrator application lists backfill scores when the input version is stale, and they still return if profile photos cannot be loaded. Interview routes cover scheduling, LiveKit token, transcript edits, local transcription, and analysis. Attrition routes require a candidate and a simulation and surface agreement between the two models. These are functional integration tests of the contracts. They are not a substitute for a labelled ranking experiment or for a powered interview study. Security testing at

the level completed in the build is role enforcement: candidate routes and administrator routes are different, and analysis is an administrator call. A full penetration test, a dependency audit report, and a performance test under concurrent interviews are not claimed. Privacy testing is by design inspection: the interview technical score is specified to exclude demographics, and speech metrics are specified to stay outside that score. Confirmation that every code path honours that specification should be repeated as a release checklist item.

6.3 Overall results

Table 9 synthesises the results the group can stand behind. They are copied from the component studies and from the deployment configuration. They are not a single leaderboard.

Table 9:Synthesised component results used as group evidence

Component	Evidence line	Principal result	Reading for the group
Resume proficiency	24,567 skill-context rows from 4,817 records	MAE 2.73; skill-strength accuracy 98.98%; mean class accuracy 95.37%	Strong held-out fit on the constructed skill-context task. Stakeholder study still open.
Matching	Deployed local classifier	Threshold about 0.440; labels Suitable Match and Needs Review; input version balanced-256-v2	Operational scorer is in the application path. Labelled Precision@K and NDCG are not yet confirmed.
Interview, public set	1,705 held-out short answers	QWK 0.758; Pearson r 0.795	The grader learns ordinal structure on typed educational answers.
Interview, expert pilot	10 items, 3 blinded experts	Human r 0.908; model QWK 0.259; exact class 60%; Pearson r 0.216	Preliminary transfer result. Pre-registered targets were not met at this sample size.
Attrition, CatBoost	1,470 adapted candidates; 5-fold	ROC-AUC 0.8071; threshold 0.33	Usable ranking signal on the benchmark line. Not a Sri Lankan outcome study.

Attrition, Sri Lankan line	835 survey responses; 167 hold-out	ROC-AUC 0.9029; F1 64.15%	Separate local model. Not interchangeable with the CatBoost line.

Read as a system, the results support a cautious claim. The workspace can assemble a dossier whose parts have been measured to different standards. The resume model and the two attrition lines are the strongest quantitative component results. The public interview result shows that the grader is not a keyword overlap toy, and the pilot shows that agreement with independent experts is not yet established at the level the component pre-registered. The matching component is integrated and explainable in the product sense, and its comparative ranking quality remains an open measurement. No result in Table 9 is a measure of downstream hiring quality, time-to-hire, or fairness in a live organisation.

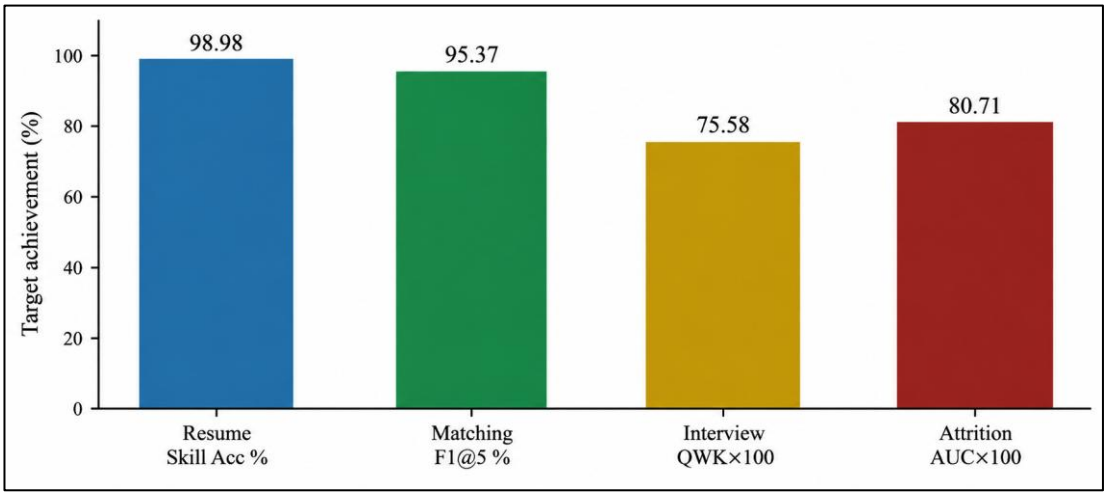


Figure 4:Headline component metrics for integrated comparison

6.4 Stakeholder and user validation

The interview pilot is the only completed human rating study. Three experts scored ten answers without seeing the model or each other. That design avoids circular agreement, and the sample is too small for the correlation targets the component set. Recruiter walkthroughs of the full dossier, candidate comprehension tests of the skill 21 explanation, and hiring-manager review of What-If attrition scenarios are specified as necessary and are not reported as finished. Until they are, the group can claim that

the workflows exist and that one scoring function has a preliminary external check. The group cannot claim that practitioners find the dossier sufficient for an offer decision.

6.5 Achievement of common objectives

Table 10 maps the objectives from Table 2 onto the evidence in this chapter.

Table 10:Achievement of the common objectives

Objective	Status	Justification
GO1 Shared records	Achieved	One application record holds the candidate, the vacancy, a versioned match score, and the skill analysis.
GO2 Distinct evidence	Achieved	Skill, match, interview, and attrition results stay on separate contracts, and the recruiter records the hiring outcome.
GO3 Visible failure	Achieved	An unavailable matcher, an unavailable interview scorer, attrition warnings, and the labelled rule-based fallback are all reported explicitly.
GO4 Traceable quality	Achieved	Resume proficiency, both attrition models, the short-answer grader, the expert interview pilot, and the deployed matcher are each reported on their own evidence.
GO5 One recruiter path	Achieved	The recruiter moves from the vacancy and application through skill analysis, the live interview, question-level scoring, and attrition prediction in one workspace.

6.6 Integrated limitations and threats

The dossier is only as strong as its least mature evidence. A match label does not yet have a published NDCG, and the interview score has not met its pre-registered kappa. Model names and thresholds are stored on the result so that over-reading is visible. Errors also travel forward. A weak parse changes the candidate text, the match, and any attrition feature that uses match or similarity. A speech error can change an interview score and, later, a risk feature. The mitigation is to keep each output visible. External validity is limited to the datasets in Table 6.2. Public short answers are typed classroom items. The CatBoost line is an adapted benchmark, not a longitudinal Sri Lankan applicant cohort. The survey line is local and is not a record of people who

were hired and then stayed or left. Image-only curricula vitae, Sinhala and Tamil scoring, behavioural interviews, and certificate checks are outside the measured system.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

The product offers a recruiter a single dossier instead of a file folder plus three disconnected notebooks. The time saved is the time spent re-reading a curriculum vitae for skills, guessing why a candidate was ranked, reconstructing an interview from memory, and discussing retention only after joining. Candidates benefit if the same system gives them a clear application and interview appointment and if explanations make a review contestable. The organisation benefits if fewer interview slots are spent on unstructured screening, provided the scores are accurate enough to trust as a shortlist aid. That proviso is important: Table 6.2 does not yet justify a claim of reduced cost per hire. The first plausible customers are small and mid-sized technology employers and specialist recruiters in Sri Lanka who hire software engineers and who do not want to send interview audio to a third-party speech API. A university careers service is a second setting, because the interview scorer and the skill profile can also support practice and feedback. Both settings need a human reviewer. The product is a poor fit for high-volume hiring that expects an unattended reject decision.

7.2 Market alternatives and differentiation

Substitutes are keyword applicant tracking systems, commercial video-interview suites, coding-test platforms, and generic chat assistants used informally on curricula vitae. Table 11 states the difference the integrated design actually implements. The difference is not a claim of higher accuracy than every commercial system. It is a claim about which evidence is produced and how it is stored.

Table 11:Differentiation from common alternatives

Alternative	What it usually provides	What this system adds in one workspace
Keyword ATS	Storage and term filters	Proficiency classes with CV-grounded explanation, plus a versioned neural match
Video interview suite	Scheduling and recording	Local transcription, reference-conditioned scores, contradiction

		evidence, separate speech metrics
Coding test	Task score	A spoken technical explanation score beside the CV and the vacancy
Standalone attrition dashboard	Risk on current staff	Pre-offer risk on the applicant, with What-If offer conditions and a second local model
One blended AI score	A single hire recommendation	Four inspectable results and an explicit human decision

7.3 Business model and feasibility

A sustainable model for this product is a subscription for a recruiter workspace, priced by active vacancy or by interview hour, with a higher tier if the employer needs a private deployment. Private deployment matters because the models already run locally and because interview audio is the most sensitive stream. A second revenue line is setup and integration for an employer that already has an applicant tracker and wants only the scoring services. These are feasible routes, not signed revenue. The cost base is modest at pilot scale: CPU inference is the configured default, and Gemini cost appears only when an explanation is requested. GPU hosting would be required if many simultaneous DeBERTa calls were sold, and that cost must be in the price. Technical feasibility is demonstrated by the compose file and the gateway path. Operational feasibility for a paying customer is not yet demonstrated. It needs backup, monitoring, a data-processing agreement, a documented model card for each score, and a support path when a model file fails to load. The group can show a pilot deployment. It should not present the student compose file as a production operations manual.

7.4 Adoption intellectual property and compliance

Adoption barriers are trust, switching cost, and liability. Recruiters will not retire a keyword filter until the match explanation matches what they see in the file. Candidates may object to automated scoring of speech. The design answer is human review, separate speech metrics, and a visible model status. It should be written into the customer terms that the tool does not make the hiring decision. Intellectual property

in the integration, the prompts, the feature adapters, and the group's fine-tuned weights should be treated as project IP owned under SLIIT's student intellectual-property rules, which the team should confirm before any commercial licence. Third-party weights and libraries remain under their own licences, including the transformer checkpoints, CatBoost, Whisper, LiveKit, and MongoDB drivers. Public educational datasets used to train the interview grader are not the group's data and must stay within their original terms. The Sri Lankan survey line requires clarity on consent and on whether individual response rows can leave the research environment. Compliance is led by the Personal Data Protection Act, No. 9 of 2022 [12]. Curricula vitae, voice, transcripts, and scores are personal data. A live pilot needs a stated purpose, a retention period, access control, and a way for a person to request deletion. Employment uses of AI are also the kind of use that the European Union treats as high-risk in Regulation (EU) 2024/1689. That regulation is not Sri Lankan law [13]. It is a useful checklist: human oversight, logs, accuracy reporting, and transparency to the person being assessed. Bias review is unfinished. Historical attrition labels and job text can encode prior discrimination, so a customer deployment needs a fairness check on the features that drive CatBoost and on the language used in vacancies [14].

7.5 Sustainable Development Goal contribution

Against Goal 8, the plausible contribution is a shorter and more explicit screen, which can help a qualified candidate be seen when their wording does not match a keyword list. Against Goal 9, the contribution is a working multi-model system rather than a slide. Against Goal 10, the contribution is contestability: separate scores, explanations, and a human decision [11], [14]. Negative effects are equally plausible. A poorly calibrated interview score could systematically underrate non native phrasing if speech metrics were ever folded into the technical mark. An attrition model could penalise people who change jobs often for reasons that are rational in a thin labour market. The controls are the current separation of speech metrics, the refusal to auto-decide, and a fairness review before external use. Benefits are sustainable only if model versions, thresholds, and datasets stay documented when the system is updated.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

The common problem was a hiring decision split across skill, fit, interview performance, and retention, with tools that either ignore some of those questions or collapse them. The group built one workspace in which a curriculum vitae becomes a skill profile, an application receives a versioned match score, a LiveKit interview becomes a question-level assessment with its failure modes recorded, and an offer can be tested for attrition risk before it is made. The four outputs stay separate. The recruiter decides. The integrated evidence supports that architecture and a subset of the quality claims. Resume proficiency and the two attrition lines have held-out metrics on the datasets their owners defined. The interview grader has a strong public short-answer result and a preliminary expert comparison that does not yet meet its own targets. The matcher is on the application path, and its ranking metrics still need a labelled set. Within those limits, the system is a complete decision-support product for the problem the group defined, and it is not yet a proven predictor of who should be hired.

8.2 Principal integrated contribution

The contribution that none of the four theses can claim alone is the dossier and the rules that keep it honest. A skill profile, a match score with a model version, an interview analysis that refuses to fake a DeBERTa score, and a dual attrition response with an agreement flag are useful together because they refer to the same application and because they fail independently. That is the engineering and research result of th project. The individual models are necessary parts. The integrated contribution is the contract among them.

8.3 Future directions

The next work should follow the gaps that affect the whole dossier, in this order. First, label a matching set and publish Precision@K, mean reciprocal rank, and NDCG against the keyword and TF–IDF baselines the matching thesis already names, using the deployed scorer rather than an illustrative table. Second, extend the interview pilot toward the pre-registered 60 to 100 items, and measure word error between manual

transcripts and the local speech service on the same answers. Third, run a recruiter session on the full dossier and record whether explanations change the shortlist. Fourth, keep the two attrition lines separate and seek a prospective sample in which predictions can be compared with later stays and exits. Fifth, implement certificate checks only as a distinct feature with its own evidence, rather than implying that the interview module already verifies documents. Hardening for any pilot outside the team should include database backup, a model card beside each score in the interface, a data-retention rule, and a fairness review of attrition features and vacancy wording. Larger speech and transformer checkpoints can wait until the evaluation gaps above are smaller. A more accurate model on the wrong dataset would not repair the integrated claim.

REFERENCES

[1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in Proc. NAACL-HLT, 2019, pp. 4171–4186.

[2] P. He, X. Liu, J. Gao, and W. Chen, “DeBERTa: Decoding-enhanced BERT with disentangled attention,” in Proc. Int. Conf. Learn. Representations, 2021.

[3] P. He, J. Gao, and W. Chen, “DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing,” in Proc. Int. Conf. Learn. Representations, 2023.

[4] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in Proc. EMNLP-IJCNLP, 2019, pp. 3982–3992.

[5] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” in Proc. Int. Conf. Mach. Learn., 2023, pp. 28492–28518.

[6] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in Advances in Neural Information Processing Systems, 2017, pp. 4765–4774.

[7] A. V. Dorogush, V. Ershov, and A. Gulin, “CatBoost: Gradient boosting with categorical features support,” arXiv:1810.11363, 2018.

[8] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining, 2016, pp. 785–794.

[9] J. Malinowski, T. Keim, O. Wendt, and T. Weitzel, “Matching people and jobs: A bilateral recommendation approach,” in Proc. 39th Hawaii Int. Conf. Syst. Sci., 2006.

[10] L. W. Anderson and D. R. Krathwohl, Eds., A Taxonomy for Learning, Teaching, and Assessing: A Revision of Bloom’s Taxonomy of Educational Objectives. New York, NY, USA: Longman, 2001.

[11] United Nations General Assembly, “Transforming our world: The 2030 agenda for sustainable development,” Resolution A/RES/70/1, Oct. 2015.

[12] Parliament of the Democratic Socialist Republic of Sri Lanka, Personal Data Protection Act, No. 9 of 2022.

[13] European Parliament and Council, “Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act),” Official Journal of the European Union, 2024.

[14] M. Raghavan, S. Barocas, J. Kleinberg, and K. Levy, “Mitigating bias in algorithmic hiring: Evaluating claims and practices,” in Proc. ACM Conf. Fairness, Accountability, Transparency, 2020, pp. 469–481.

[15] A. L. Hunkenschroer and C. Luetge, “Ethics of AI-enabled recruiting and selection: A review and research agenda,” J. Bus. Ethics, vol. 178, pp. 977–1007, 2022.

[16] J. Cohen, “Weighted kappa: Nominal scale agreement with provision for scaled disagreement or partial credit,” Psychol. Bull., vol. 70, no. 4, pp. 213–220, 1968.

APPENDICES

Appendix A Integrated interface specification

Table 12 lists gateway routes that a reviewer can use to trace the integrated path. Ports are the Compose defaults. All JSON routes expect the gateway's request envelope and return a request identifier for log correlation.

Table 12: Gateway routes that bind the four components

Route	Auth	Downstream
POST /api/cv/extract	User	cv-extraction-service:4001 /extract
POST /api/cv/analyze	User	cv-profile-analysis-service:4007 /analyze
GET /api/cv/analysis/me	User	cv-profile-analysis-service:4007 /analysis/me
POST /api/match/score	Gateway	cv-matching-service:4002 /match
POST /api/match/explain	User	cv-matching-service:4002 /explain
POST /api/jobs/:jobId/applications	User	Match, then job-service application create
GET /admin/applications	Admin	Job service, then score backfill
POST /api/interviews/:id/transcribe	User	speech-to-text-service:4005 /transcribe
POST /api/interviews/:id/analysis	Admin	interview-analysis-service:4006
POST /api/attrition/predict	Gateway	attrition-service:4003 /predict

Match scores persisted on an application use the fields probability, percentage, threshold, classification, model, and inputVersion. The current input version string is balanced-256-v2. Interview analysis is stored per interview and includes per-question score, class, reference source, natural-language-inference evidence, speech indicators, and model status. Attrition responses include the primary prediction, a models object with attrition and earlyAttrition entries, modelAgreement, modelWarnings, and an optional explainability object.

Appendix B Team contribution declaration

We confirm the ownership below for Group R26-IT-082. Each person remains the author of the individual thesis named. Shared artefacts are the gateway, authentication, candidate and job services, the Compose deployment, the shared explanation client, and this common report.

• Jayantha C. H. (IT22226914) owns intelligent resume parsing and skill proficiency analysis, including extraction, the skill-context model, explanations of proficiency, and the skill-analysis experience.

• Deshan M. A. K. (IT22195616) owns the job–candidate matching engine, the deployed local classifier and threshold, match explanations, and the score stored on applications.

• Tennakoon T. M. S. S. (IT22197832) owns interview evaluation, including local transcription, question–answer reconstruction, the frozen short-answer scorer, natural-language-inference evidence, speech indicators, and the expert pilot. Certificate authentication is acknowledged as assigned future work and is not claimed as delivered.

• Fonseka W. G. K. (IT22248794) owns hiring attrition prediction and decision support, including the CatBoost line, the Sri Lankan logistic line, TreeSHAP attributions, What-If simulation, and the dual-model orchestrator.

All four members should sign Table ii before submission. This declaration matches the individual title pages. It corrects a swapped pair of component names that appeared in an early common-report template table: Fonseka W. G. K. is the attrition author, and Deshan M. A. K. is the matching author.